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United States Patent Application Publication

20250271049

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Finke; Aaron M.

BIDIRECTIONAL TO UNIDIRECTIONAL GEARSET

Abstract

An apparatus includes a gearset that includes an input gear configured to be rotated by an input shaft, first and second one-way clutches, first and second output gears configured to drive rotation of an output shaft via the first and second one-way clutches, respectively, and an idler gear. The second output gear has a smaller outer diameter than the first output gear. The input gear is configured to (i) rotate the first output gear in a first direction and (ii) rotate the idler gear in the first direction. The idler gear is configured to rotate the second output gear in a second direction opposite the first direction. The first and second one-way clutches are configured to allow the first and second output gears to drive rotation of the output shaft in a single predetermined rotational direction regardless of a rotational direction of the input shaft.

Inventors: Finke; Aaron M. (Janesville, WI)

Applicant: Hamilton Sundstrand Corporation (Charlotte, NC)

Family ID: 1000007728063

Appl. No.: 18/588903

Filed: February 27, 2024

Publication Classification

Int. Cl.: F16H1/00 (20060101); F16H1/20 (20060101); F16H1/22 (20060101); H02K7/116 (20060101); H02K9/193 (20060101)

U.S. Cl.:

CPC F16H1/003 (20130101); F16H1/206 (20130101); F16H1/22 (20130101); H02K7/116 (20130101); H02K9/193 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] This disclosure is generally directed to gearboxes and related systems and processes. More specifically, this disclosure is directed to a bidirectional to unidirectional gearset.

BACKGROUND

[0002] Turbomachines and other mechanical devices often include rotating input shafts that are used to drive rotating output shafts. The conversion of input rotation from an input shaft into output rotation of an output shaft is typically accomplished using a set of gears, which is referred to as a gearset.

SUMMARY

[0003] This disclosure is directed to a bidirectional to unidirectional gearset.

[0004] In a first embodiment, a system includes a rotatable shaft having a first end and an opposing second end, a first mechanical device configured to receive a first rotational input from the first end of the rotatable shaft, and a second mechanical device configured to receive a second rotational input from the second end of the rotatable shaft. The second rotational input is opposite the first rotational input. Each of the first and second mechanical devices includes a gearset that includes an input gear configured to be rotated by the rotatable shaft, first and second one-way clutches, first and second output gears configured to drive rotation of an output shaft via the first and second one-way clutches, respectively, and an idler gear. The input gear of the first gearset is configured to (i) rotate the first output gear of the first gearset in a first direction and (ii) rotate the idler gear of the first gearset in the first direction, and the idler gear of the first gearset is configured to rotate the second output gear of the first gearset in a second direction opposite the first direction. The input gear of the second gearset is configured to (i) rotate the first output gear of the second gearset in the second direction and (ii) rotate the idler gear of the second gearset in the second direction, and the idler gear of the second gearset is configured to rotate the first output gear of the second gearset in the first direction. The first and second one-way clutches of each gearset are configured to allow the first and second output gears of the gearset to drive rotation of the associated output shaft in a single predetermined rotational direction regardless of a rotational direction of the rotatable shaft relative to the gearset.

[0005] In a second embodiment, an apparatus includes a gearset that includes an input gear configured to be rotated by an input shaft, first and second one-way clutches, first and second output gears configured to drive rotation of an output shaft via the first and second one-way clutches, respectively, and an idler gear. The second output gear has a smaller outer diameter than the first output gear. The input gear is configured to (i) rotate the first output gear in a first direction and (ii) rotate the idler gear in the first direction. The idler gear is configured to rotate the second output gear in a second direction opposite the first direction. The first and second one-way clutches are configured to allow the first and second output gears to drive rotation of the output shaft in a single predetermined rotational direction regardless of a rotational direction of the input shaft.

[0006] In a third embodiment, a method includes imparting rotational input to an input gear of a gearset using an input shaft. The method also includes driving rotation of an output shaft using first and second output gears of the gearset respectively via first and second one-way clutches and an idler gear of the gearset. The input gear (i) rotates the first output gear in a first direction and (ii) rotates the idler gear in the first direction. The idler gear rotates the second output gear in a second direction opposite the first direction. The first and second one-way clutches allow the first and second output gears to drive rotation of the output shaft in a single predetermined rotational direction regardless of a rotational direction of the input shaft. The second output gear has a smaller outer diameter than the first output gear.

[0007] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a more complete understanding of this disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

[0009] FIGS. 1 and 2 illustrate an example bidirectional to unidirectional gearset according to this disclosure;

[0010] FIGS. 3 and 4 illustrate example operations of a bidirectional to unidirectional gearset according to this disclosure; and

[0011] FIGS. 5 through 7 illustrate an example system that incorporates multiple bidirectional to unidirectional gearsets according to this disclosure.

DETAILED DESCRIPTION

[0012] FIGS. 1 through 7, described below, and the various embodiments used to describe the principles of the present disclosure are by way of illustration only and should not be construed in any way to limit the scope of this disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any type of suitably arranged device or system.

[0013] As described above, turbomachines and other mechanical devices often include rotating input shafts that are used to drive rotating output shafts. The conversion of input rotation from an input shaft into output rotation of an output shaft is typically accomplished using a set of gears, which is referred to as a gearset. Unfortunately, turbomachines and other mechanical devices are generally designed to receive input rotation in a specific direction, which is typically expressed as either clockwise (CW) rotation or counterclockwise (CCW) rotation. In some situations, due to space constraints or other constraints, it is not possible to provide a desired direction of rotation to a turbomachine or other mechanical device. In other situations, there may be a desire to use multiple turbomachines or other mechanical devices, but the direction of rotation may be reversed at the locations of the turbomachines or other mechanical devices. One approach to solving the latter issue may be to design different versions of the turbomachines or other mechanical devices that accept input rotations in different directions, but this can lead to inefficiencies in design and manufacture, lead to installation mistakes, and require personnel to maintain different devices in stock for use.

[0014] This disclosure provides bidirectional to unidirectional gearsets that have the ability to convert bidirectional input rotation into unidirectional output rotation. In other words, a bidirectional to unidirectional gearset generates output rotation in the same direction (one of clockwise or counterclockwise) regardless of whether the input rotation is clockwise or counterclockwise. As described in more detail below, the gearset includes an input gear, first and second output gears, and an idler gear. The input gear can be coupled to an input shaft, and the first and second output gears can both be coupled to an output shaft. The input gear is configured to simultaneously drive the first output gear and the idler gear, and the idler gear is configured to drive the second output gear. One-way clutches are positioned between the first and second output gears and the output shaft, and the one-way clutches allow the first or second output gear to drive rotation of the output shaft in the same direction. During operation, the first output gear may be used to drive the output shaft when the input shaft is rotating in one direction, and the second output gear may be used to drive the output shaft when the input shaft is rotating in the opposite direction. In this way, the output shaft rotates in the same direction regardless of the direction of the input rotation.

[0015] This disclosure also provides systems that can use multiple instances of such a bidirectional to unidirectional gearset. For example, a system may include multiple mechanical devices, such as multiple turbomachines like electric generators with integral or coupled oil pumps. The multiple

mechanical devices can be coupled to opposite ends of a rotating input shaft. As a result, the mechanical devices can receive input rotation in different directions, such as when one mechanical device at one end of the rotating input shaft receives clockwise input rotation and another mechanical device at the opposite end of the rotating input shaft receives counterclockwise input rotation. Each mechanical device can include an instance of the bidirectional to unidirectional gearset, which allows each mechanical device to convert its respective input rotation into a suitable output rotation for use by the mechanical device or other device. In this way, a common design can be used for the mechanical devices, rather than separate designs for mechanical devices used at different locations.

[0016] FIGS. **1** and **2** illustrate an example bidirectional to unidirectional gearset **100** according to this disclosure. As shown in FIGS. **1** and **2**, the gearset **100** includes an input gear **102**, which can be coupled to and rotate with an input shaft **104**. The input shaft **104** represents a structure that can provide input rotation to the gearset **100**. As shown here, the input shaft **104** may rotate clockwise and/or counterclockwise. The exact rotation direction of the input shaft **104** can vary depending on the application. In some embodiments, for example, the input shaft **104** may rotate in only one direction (either clockwise or counterclockwise), such as when the input shaft **104** is designed or is coupled to another component that is designed to rotate in only one direction. In other embodiments, the rotation direction of the input shaft **104** may vary over time.

[0017] The gearset **100** also includes a first output gear **106** and a second output gear **108**, each of which can be coupled to and rotate about an output shaft **110**. The output shaft **110** represents a structure that can provide output rotation from the gearset **100**, such as to an external component that is designed to receive rotation in a specific direction. As shown here, the output shaft **110** may rotate in only one direction, either clockwise or counterclockwise. Again, the exact rotation direction of the output shaft **110** can vary depending on the application. The gearset **100** further includes an idler gear **112**, which can be connected to and rotate with an idler shaft **114**. The idler shaft **114** represents a structure that can allow the idler gear **112** to rotate freely in either direction.

[0018] The output gears **106** and **108** are respectively connected to the output shaft **110** using one-way clutches **116** and **118**. Each one-way clutch **116** and **118** represents a structure that allows the associated output gear **106** and **108** to drive rotation of the output shaft **110** in one direction and prevent rotation of the output shaft **110** in the opposite direction. In other words, the one-way clutches **116** and **118** are configured to engage and allow the output shaft **110** to be respectively driven by the output gears **106** and **108** in one direction and to disengage and prevent the output shaft **110** from being respectively driven by the output gears **106** and **108** in the opposite direction. For example, the one-way clutch **116** may engage and allow the first output gear **106** to drive the output shaft **110** in one direction (such as counterclockwise), in which case the one-way clutch **118** may disengage and allow the second output gear **108** to rotate freely without driving the output shaft **110**. Also, the one-way clutch **118** may engage and allow the second output gear **108** to drive the output shaft **110** in the same direction (such as counterclockwise), in which case the one-way clutch **116** may disengage and allow the first output gear **106** to rotate freely without driving the output shaft **110**. As described below, this configuration allows for the conversion of the input rotation from the input shaft **104** (regardless of its direction) into a known single-direction output rotation of the output shaft **110**. Each one-way clutch **116** and **118** may represent any suitable structure configured to allow selective driving of the output shaft **110** depending on rotation direction, such as a sprag, roller, pawl, or other type of one-way clutch.

[0019] Each gear **102**, **106**, **108**, **112** includes any suitable structure having teeth or other elements that can interact with and rotate or be rotated by another gear. Each gear **102**, **106**, **108**, **112** may be formed using any suitable material(s), such as one or more metals. Each gear **102**, **106**, **108**, **112** may also be formed using any suitable technique(s), such as an additive or subtractive manufacturing process. In addition, each gear **102**, **106**, **108**, **112** may have any suitable size, shape, and dimensions.

[0020] In some embodiments, such as the one shown in FIGS. 1 and 2, the input gear **102** can have a substantially uniform outer diameter, at least at the portions of the input gear **102** that contact the first output gear **106** and the idler gear **112**. Also, in some embodiments, the first and second output gears **106** and **108** can have different outer diameters, such as when the first output gear **106** has a larger outer diameter than the second output gear **108**. Among other things, this can allow the idler gear **112** to be positioned partially over the first output gear **106** in the view shown in FIG. 1, which may help to reduce the overall package size of the gearset **100**. That is, when the gearset **100** is viewed from a direction parallel to the axes of the input and output shafts **104** and **110**, a portion of the idler gear **112** can overlap a portion of the first output gear **106**. However, the actual sizes of the gears **102**, **106**, **108**, **112** can vary based on a number of factors.

[0021] FIGS. 3 and 4 illustrate example operations of the bidirectional to unidirectional gearset **100** according to this disclosure. As shown in FIG. 3, the input shaft **104** is rotating clockwise, which causes the input gear **102** to also rotate clockwise. This clockwise rotation of the input gear **102** causes counterclockwise rotation of the first output gear **106**. This clockwise rotation of the input gear **102** also causes counterclockwise rotation of the idler gear **112**, which causes the second output gear **108** to rotate clockwise. The opposite rotational directions of the output gears **106** and **108** (which are coupled to the same output shaft **110**) are accommodated using the one-way clutches **116** and **118**. More specifically, the one-way clutch **116** is engaged and the one-way clutch **118** is disengaged here, which allows the first output gear **106** to rotate the output shaft **110** counterclockwise. The second output gear **108** may continue to rotate clockwise, but the one-way clutch **118** prevents the second output gear **108** from driving rotation of the output shaft **110**.

[0022] As shown in FIG. 4, the input shaft **104** is rotating counterclockwise, which

[0023] causes the input gear **102** to also rotate counterclockwise. This counterclockwise rotation of the input gear **102** causes clockwise rotation of the first output gear **106**. This counterclockwise rotation of the input gear **102** also causes clockwise rotation of the idler gear **112**, which causes the second output gear **108** to rotate counterclockwise. Again, the opposite rotational directions of the output gears **106** and **108** (which are coupled to the same output shaft **110**) are accommodated using the one-way clutches **116** and **118**. More specifically, the one-way clutch **118** is engaged and the one-way clutch **116** is disengaged here, which allows the second output gear **108** to rotate the output shaft **110** counterclockwise. The first output gear **106** may continue to rotate clockwise, but the one-way clutch **116** prevents the first output gear **106** from driving rotation of the output shaft **110**.

[0024] As a result, the output shaft **110** can rotate counterclockwise regardless of the rotational direction of the input shaft **104**. Note, however, that the rotational direction of the output shaft **110** may be clockwise using suitable changes to the design of the gearset **100**, such as by reversing the directions in which the one-way clutches **116** and **118** engage and disengage.

[0025] Although FIGS. 1 and 2 illustrate one example of a bidirectional to unidirectional gearset **100** and FIGS. 3 and 4 illustrate examples of operations of the bidirectional to unidirectional gearset **100**, various changes may be made to FIGS. 1 through 4. For example, the sizes of various gears **102**, **106**, **108**, **112** in the gearset **100** can be tuned or adjusted in order to provide desired gear ratios within the gearset **100**. Also, the one-way clutches **116** and **118** can be configured to engage and disengage in desired directions in order to impart desired rotation to the output shaft **110**.

[0026] FIGS. 5 through 7 illustrate an example system **500** that incorporates multiple bidirectional to unidirectional gearsets **100** according to this disclosure. As shown in FIGS. 5 and 6, the system **500** includes a pair of generators **502** and **504** and an accessory gearbox **506**. The accessory gearbox **506** represents a structure used with an aircraft engine to drive various accessories associated with an aircraft or the aircraft engine. In this example, the accessory gearbox **506** includes a rotatable shaft **602**, which is used to drive the generators **502** and **504**. In some embodiments, the generators **502** and **504** may represent turbomachines, such as electric generators

with integral or coupled oil pumps. As a particular example, each of the generators **502** and **504** may include an oil pump connected to an electric generator.

[0027] As can be seen here, the shaft **602** may be designed to rotate in a single direction, which in this example is counterclockwise in the views shown in FIGS. **5** and **6**. As a result, the shaft **602** (which represents an input shaft for each generator **502** and **504**) has opposite rotational directions from the perspective of the generators **502** and **504**. That is, while looking at the generator **502** from a point of view along the shaft **602**, the shaft **602** is rotating counterclockwise with respect to the generator **502**. While looking at the generator **504** from the same point of view along the shaft **602**, the shaft **602** is rotating clockwise with respect to the generator **504**.

[0028] As noted above, one approach to dealing with these opposite rotational directions may be to use separate distinct designs for the generators **502** and **504**. However, this increases development and recurring costs and requires the use of distinct hardware. As a result, this approach could require maintaining different components in stock and could lead to mistaken installation of one generator in an unusable position.

[0029] As shown in FIG. **7**, each generator **502** and **504** can include or be coupled to an instance of the gearset **100**, where the input shaft **104** represents or is coupled to the rotatable shaft **602**. Each generator **502** and **504** in this example includes an electric generator that is formed using a rotor **702** and a stator **704**. The output shaft **110** of the associated gearset **100** can cause the rotor **702** to rotate, which causes the stator **704** to generate electrical power **706**. The output shaft **110** can also be coupled to and drive an oil pump **708**. This arrangement can be replicated in both generators **502** and **504**, which allows each generator **502** and **504** to drive its associated oil pump **708** in the same rotational direction, regardless of the generators **502** and **504** of the two input shafts **104** of the gearsets **100**.

[0030] By incorporating an instance of the gearset **100** described above into each generator **502** and **504**, the generators **502** and **504** can have a common design regardless of the direction of rotation of the shaft **602** relative to the generators **502** and **504**. As a result, the gearsets **100** in the generators **502** and **504** allow a single design to be used, which can lower development and recurring costs since it enables common hardware to be used. This can also reduce potential installation mistakes since it is immaterial which device is installed on the left and right sides of the shaft **602**. In other words, the generators **502** and **504** are interchangeable. In addition, when the generators **502** and **504** represent electric generators with integral or coupled oil pumps **708**, it becomes possible to adjust the pump flows higher or lower for different rotational directions of the shaft **602** if needed or desired. As a particular example, if one generator **502** or **504** has higher losses or is otherwise less efficient compared to the other generator **504** or **502** due to the difference in input rotational direction, the gear ratios of the less efficient generator **502** or **504** can be tuned to rotate its oil pump **708** at a higher speed (in order to compensate for its lower efficiency and provide more cooling).

[0031] Although FIGS. **5** through **7** illustrate one example of a system **500** that incorporates multiple bidirectional to unidirectional gearsets **100**, various changes may be made to FIGS. **5** through **7**. For example, one or more instances of the bidirectional to unidirectional gearset **100** may be used with any other suitable mechanical device(s) and in any other suitable system(s).

[0032] It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the

following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

[0033] The description in the present disclosure should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims invokes 35 U.S.C. § 112(f) with respect to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the features of the claims themselves, and is not intended to invoke 35 U.S.C. § 112(f).

[0034] While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. A system comprising: a rotatable shaft having a first end and an opposing second end; a first mechanical device configured to receive a first rotational input from the first end of the rotatable shaft; and a second mechanical device configured to receive a second rotational input from the second end of the rotatable shaft, the second rotational input opposite the first rotational input; wherein the first and second mechanical devices respectively comprise first and second gearsets, each of the gearsets comprising: an input gear configured to be rotated by the rotatable shaft; first and second one-way clutches; first and second output gears configured to drive rotation of an output shaft via the first and second one-way clutches, respectively; and an idler gear; wherein the input gear of the first gearset is configured to (i) rotate the first output gear of the first gearset in a first direction and (ii) rotate the idler gear of the first gearset in the first direction, and wherein the idler gear of the first gearset is configured to rotate the second output gear of the first gearset in a second direction opposite the first direction; wherein the input gear of the second gearset is configured to (i) rotate the first output gear of the second gearset in the second direction and (ii) rotate the idler gear of the second gearset in the second direction, and wherein the idler gear of the second gearset is configured to rotate the first output gear of the second gearset in the first direction; and wherein the first and second one-way clutches of each gearset are configured to allow the first and second output gears of the gearset to drive rotation of the associated output shaft in a single predetermined rotational direction regardless of a rotational direction of the rotatable shaft relative to the gearset.
2. The system of claim 1, wherein: the first and second mechanical devices comprise first and second generators; and the rotatable shaft forms part of an aircraft accessory gearbox.
3. The system of claim 1, wherein each of the first and second mechanical devices comprises: an electric generator; and an oil pump coupled to or integral with the electric generator.
4. The system of claim 1, wherein: the rotatable shaft is configured to rotate clockwise relative to the first mechanical device and counterclockwise relative to the second mechanical device; and the first and second gearsets of the first and second mechanical devices are configured to cause rotation of both of the output shafts in the single predetermined rotational direction when the rotatable shaft is rotating clockwise and when the rotatable shaft is rotating counterclockwise.
5. The system of claim 1, wherein, in each of the first and second gearsets, the second output gear of the gearset has a smaller outer diameter than the first output gear of the gearset.
6. The system of claim 5, wherein, in each of the first and second gearsets, a portion of the idler

gear of the gearset overlaps a portion of the first output gear of the gearset when the gearset is viewed from a direction parallel to an axis of the associated output shaft.

7. The system of claim 1, wherein: the first one-way clutch of each gearset is configured to engage when the first output gear of the gearset is rotating in one of the first and second directions and disengage when the first output gear of the gearset is rotating in another of the first and second directions; and the second one-way clutch of each gearset is configured to engage when the second output gear of the gearset is rotating in one of the first and second directions and disengage when the second output gear of the gearset is rotating in another of the first and second directions.

8. The system of claim 1, wherein the first and second mechanical devices are interchangeable.

9. An apparatus comprising: a gearset comprising: an input gear configured to be rotated by an input shaft; first and second one-way clutches; first and second output gears configured to drive rotation of an output shaft via the first and second one-way clutches, respectively, the second output gear having a smaller outer diameter than the first output gear; and an idler gear; wherein the input gear is configured to (i) rotate the first output gear in a first direction and (ii) rotate the idler gear in the first direction; wherein the idler gear is configured to rotate the second output gear in a second direction opposite the first direction; and wherein the first and second one-way clutches are configured to allow the first and second output gears to drive rotation of the output shaft in a single predetermined rotational direction regardless of a rotational direction of the input shaft.

10. The apparatus of claim 9, wherein: the input shaft is configured to rotate clockwise or counterclockwise; and the gearset is configured to cause rotation of the output shaft in the single predetermined rotational direction when the input shaft is rotating clockwise and when the input shaft is rotating counterclockwise.

11. The apparatus of claim 9, wherein a portion of the idler gear overlaps a portion of the first output gear when the gearset is viewed from a direction parallel to an axis of the output shaft.

12. The apparatus of claim 9, wherein: the first one-way clutch is configured to engage when the first output gear is rotating in one of the first and second directions and disengage when the first output gear is rotating in another of the first and second directions; and the second one-way clutch is configured to engage when the second output gear is rotating in one of the first and second directions and disengage when the second output gear is rotating in another of the first and second directions.

13. The apparatus of claim 9, wherein: the gearset forms part of or is coupled to a generator; and the generator is coupled to a rotatable shaft forming part of an aircraft accessory gearbox.

14. The apparatus of claim 13, wherein the gearset is configured to cause rotation of the output gear in the single predetermined rotational direction regardless of whether the generator receives a first rotational input from a first end of the rotatable shaft or a second rotational input from a second end of the rotatable shaft.

15. The apparatus of claim 9, wherein: the gearset forms part of or is coupled to a generator; the generator comprises: an electric generator; and an oil pump coupled to or integral with the electric generator; and the output shaft is configured to drive the electric generator and the oil pump.

16. A method comprising: imparting rotational input to an input gear of a gearset using an input shaft; and driving rotation of an output shaft using first and second output gears of the gearset respectively via first and second one-way clutches and an idler gear of the gearset; wherein the input gear (i) rotates the first output gear in a first direction and (ii) rotates the idler gear in the first direction; wherein the idler gear rotates the second output gear in a second direction opposite the first direction; wherein the first and second one-way clutches allow the first and second output gears to drive rotation of the output shaft in a single predetermined rotational direction regardless of a rotational direction of the input shaft; and wherein the second output gear has a smaller outer diameter than the first output gear.

17. The method of claim 16, wherein: the gearset represents a first gearset; and the method further comprises: imparting rotational input to an input gear of a second gearset, the rotational input to the

input gear of the second gearset opposite the rotational input to the input gear of the first gearset; and driving rotation of another output shaft using first and second output gears of the second gearset respectively via first and second one-way clutches and an idler gear of the second gearset.

18. The method of claim 16, wherein: the gearset forms part of or is coupled to a generator; and imparting the rotational input to the input gear comprises imparting the rotational input to the input gear based on rotation of a rotatable shaft forming part of an aircraft accessory gearbox.

19. The method of claim 18, wherein: the generator comprises an electric generator and an oil pump coupled to or integral with the electric generator; and the output shaft drives the electric generator and the oil pump.

20. The method of claim 16, wherein: the input shaft is configured to rotate clockwise or counterclockwise; and the gearset causes rotation of the output shaft in the single predetermined rotational direction when the input shaft is rotating clockwise and when the input shaft is rotating counterclockwise.
